MILITARY SPECIFICATION

INVERTER

(FOR TORPEDO MARK 37 MODS 0, 1, AND 3)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification establishes the requirements for the procurement of an inverter, Drawing 1263725, that transforms 26 volt dc battery voltage into regulated 3-phase, 400 Hertz (Hz), ac voltage of 115 volts average of the three line-to-line voltages.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-Z-325	Zinc Coating, Electrodeposited; Requirements for

Military

MIL-P-116	Preservation, Methods of
MIL-M-3171	Magnesium Alloy; Processes for Corrosion Protection for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys

FSC 1355

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MIL-P-6808	Primer Coating, Zinc Chromate, for Aircraft Use
MIL-W-16878	Wire, Electrical, Insulated, High Temperature
MIL-S-19500	Semi-Conductor Devices; General Specification for
MIL-E-82590	Environmental Requirements and Tests (For Torpedo Mk 37 Components and Assemblies)

STANDARDS

Military

MIL-STD-147	Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners or Pallet-Type Base

DRAWINGS

Naval Ordnance Systems Command

Dwg 1263725	Inverter
LD 495963	Shipping (Packing) Condition
Dwg 1803667	Shipping (Packing) Condition

PUBLICATIONS

Naval Ordnance Systems Command

OS 6341	Miscellaneous General Ordnance Design Requirements

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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National Bureau of Standards Publications

Handbook H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Description. The inverter is a device which transforms the torpedo dc battery voltage into a regulated three-phase, ac voltage for use by the torpedo control system. It consists of a 3-phase, 400 Hz, 115 volt root-mean-square (rms) alternator contained in a common housing with a 24 volt dc driver motor. The automatic output voltage and frequency regulators are completely solid state devices integrally mounted in the unit. Externally accessible manual setting means for adjusting output frequency and voltage are provided. All electrical connections to the inverter are made through a standard MS type receptacle.

3.2 General requirements

3.2.1 The inverter shall conform in all respects to Drawing 1263725.

3.2.2 Data required. No data is required by this specification (other than reports accompanying samples submitted for preproduction and periodic production testing), or by applicable documents referenced in Section 2, unless specified in the contract or order (see 6.2).

3.2.3 Preproduction sample. Unless otherwise specified in the contract or order, preproduction samples of the inverter shall be manufactured using the methods and procedures proposed for the production lot. These units shall be tested as specified in Section 4 herein and are for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing inverters, that comply with the technical requirements of the contract. See 4.3.1.

3.2.4 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the inverter shall. conform strictly to the detailed requirements indicated on Dwg 1263725.

3.2.4.1 Corrosion resistance. Materials shall be of a corrosion-resisting type, or shall be suitably processed to resist corrosion.

3.2.4.2 Standard parts and commercial items. The greatest practicable use commensurate with design requirements shall be made of standard parts and commercial items.

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3.2.4.3 Hookup wire. Unless otherwise specified, all hookup wire used in the inverter shall conform to MIL-W-16878.

3.2.4.4 Mercury and silicone. Materials capable of giving off either mercury or silicone vapor when subjected to temperature within the operating range of the inverter shall not be used.

3.2.4.5 Ferrous metal parts. Electrical sheet steel assemblies shall be processed by accepted commercial methods to provide protection against corrosion. Where other non-corrosive-resistant ferrous metals must be used, the exposed surfaces shall be zinc plated in accordance with QQ-Z-325, Type II, Class 3 (0.002 inch thick zinc plating.)

3.2.4.6 Aluminum alloy parts. Aluminum and aluminum alloys, if used, shall be coated to meet the requirements of MIL-C-5541.

3.2.4.7 Magnesium alloy parts. Magnesium and magnesium alloys, if used, shall be treated in accordance with MIL-M-3171, Type III, followed by priming in accordance with MIL-P-6808.

3.2.4.8 Semi-conductor devices. All semi-conductor devices shall meet the requirements of MIL-S-19500. Prior to use, these devices shall be subjected to 100 percent rated power dissipation for 200 hours and thereafter, meet the initial acceptance requirements.

3.2.4.9 Unauthorized components. No electrolytic capacitor shall be used in this inverter.

3.2.4.10 Weight. The weight of the inverter shall not exceed 23 pounds.

3.2.5 Moisture, fungus, and corrosion. The inverter shall be rendered moisture, fungus, and corrosion resistant as required by OS 6341.

3.2.6 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with Dwg 1263725.

3.2.7 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

3.2.8 Modifications. Any change(s), after initial qualification, of schematic, materials, components, and processes shall be approved in writing by the procuring activity.

3.2.9 Conflicting requirements. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall apply.

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3.3 Performance requirements and product characteristics. The inverter shall meet the following performance requirements and product characteristics:

3.3.1 Performance requirements

3.3.1.1 Input

3.3.1.1.1 Source voltage. The nominal steady state source voltage shall be within 26 plus or minus 2 volts dc open circuit.

3.3.1.1.2 Circuit impedance. The total impedance looking back from the inverter terminals to the source voltage shall be within 0.07 to 0.09 ohm resistive, inclusive of all metering resistances. This impedance shall apply to all tests.

3.3.1.1.3 Peak starting current. The peak current shall not exceed 200 amperes, with 98 plus or minus 2 percent rated load connected to the inverter at a source voltage of 28 plus or minus 0.2 volts dc.

3.3.1.1.3.1 Decay time. Within 1.0 second after starting the inverter, the input current shall be less than the arithmetic average of the peak starting current and the final running current.

3.3.1.2 Starting conditions

3.3.1.2.1 Increasing voltage. The starting voltage as measured at the source terminals shall not deviate by more than 2 volts peak from the following curve: $V = 10 + 1.6t$ where V = dc input volts
t = time in seconds measured from turn on

until a steady state value of 26 plus or minus 2 volts dc is reached with a load of 62 plus or minus 2 percent of rated load.

3.3.1.2.2 Step voltage. The inverter will be subjected to the following step-voltage conditions:

3.3.1.2.2.1 Open circuit source voltage prior to starting at 24 plus or minus 0.2 volts dc; inverter load at 98 plus or minus 2 percent of rated load.

3.3.1.2.2.2 Open circuit source voltage prior to starting at 24 plus or minus 0.2 volts dc; inverter load at 62 plus or minus 2 percent of rated load.

3.3.1.2.2.3 Open circuit source voltage prior to starting at 28 plus

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or minus 0.2 volts dc; inverter load at 98 plus or minus 2 percent of rated load.

3.3.1.2.2.4 Open circuit source voltage prior to starting at 28 plus or minus 0.2 volts dc; inverter load at 62 plus or minus 2 percent of rated load.

3.3.1.3 Output

3.3.1.3.1 Rated load. The rated output load shall be 3-phase, 550 volt-amperes, 64 percent lagging power factor, balanced delta, at 115 volts root-mean-square (rms), 400 Hz.

3.3.1.3.2 Voltage. The three regulated line-to-line voltages shall be within, and remain within, 115 plus or minus 1 volts rms, with the difference between any two line-to-line voltages not to exceed 1 volt rms. The output voltage may be reset to hold within the specified limits once during each 10 hours of operation.

3.3.1.3.2.1 Peak voltage. The instantaneous voltage shall not exceed plus or minus 250 volts peak.

3.3.1.3.3 Frequency. The frequency shall be 400 plus or minus 4 Hz. The frequency may be reset to hold within the specified limits once during each 10 hours of operation.

3.3.1.3.4 Regulation. The output voltage measured at the inverter terminals shall be within the limits of 3.3.1.3.2 and the output frequency shall be within the limits of 3.3.1.3.3, when the inverter is subjected to the following conditions:

3.3.1.3.4.1 Increasing voltage as specified in 3.3.1.2.1 with 62 plus or minus 2 percent rated load and after a steady state value over the range of 24 to 28 volts dc is reached.

3.3.1.3.4.2 Step-voltage as specified in the conditions of 3.3.1.2.2.

3.3.1.3.5 Phase sequence. The phase sequence of the output voltage, with the inverter terminals connected in conformance with 3.3.2.4.3, shall be ABC.

3.3.1.3.6 Harmonic voltage content

3.3.1.3.6.1 Total harmonic voltage. The total harmonic voltage content to 10 kiloHertz (kHz) shall not exceed 2 percent of the fundamental output voltage.

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3.3.1.3.6.2 Individual harmonic voltages. All harmonic voltages except the 7th shall not exceed 1 percent of the fundamental output voltage. The 7th harmonic voltage shall not exceed 1.5 percent of the fundamental output voltage.

3.3.1.3.7 Modulation. Output voltage modulation shall not exceed 1 percent of the output voltage when the inverter is operating in accordance with 3.3.1.2.2.

3.3.1.3.8 Response time

3.3.1.3.8.1 Step-voltage start. At 3 seconds maximum after the inverter is started under the conditions of 3.3.1.2.2, the output voltage, phase balance, and frequency shall be within, and remain within, the limits of 3.3.1.3.2 and 3.3.1.3.3.

3.3.1.3.8.2 Increasing voltage start. Within 11 seconds maximum after the inverter is started in accordance with 3.3.1.2.1, the output voltage, phase balance, and frequency shall be within, and remain within, the limits of 3.3.1.3.2 and 3.3.1.3.3.

3.3.1.3.9 Recovery time. At 0.1 second maximum after the inverter load is instantaneously switched from 62 plus or minus 2 percent to 98 plus or minus 2 percent of rated load, the AB phase voltage shall be within, and remain within, 115 plus or minus 1 volts rms; the output frequency shall be within, and remain within, 400 plus or minus 4 Hz. These requirements shall be met under the input requirements of 3.3.1.1.

3.3.1.4 Efficiency. The minimum efficiency of the inverter, at 98 plus or minus 2 percent rated load, shall be not less than 40 percent. The minimum efficiency of the inverter at 62 plus or minus 2 percent rated load shall be not less than 35 percent.

3.3.1.5 Commutation. Commutation shall be such that not more than fine pinpoint sparking occurs when the inverter is operated at 98 plus or minus 2 percent and 62 plus or minus 2 percent of rated load.

3.3.1.6 Dielectric requirements

3.3.1.6.1 DC circuits. All dc circuitry of the inverter must be capable of withstanding 500 volts rms, 60 Hz to the frame, with the ac circuits grounded to the frame, for a period of at least 1 minute without any arcing or breakdown of insulation.

3.3.1.6.1.1 The initiation resistance at 500 volts dc shall be 100 megohms minimum between all inverter dc circuits and the frame, with all ac circuits grounded to the frame.

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3.3.1.6.1.1.1 Within 3 hours of completion of the temperature and humidity nonoperating test, the insulation resistance shall be 1 megohm minimum.

3.3.1.6.1.1.2 Within 48 hours of completion of the temperature and humidity nonoperating test, the insulation resistance shall be 100 megohms minimum.

3.3.1.6.1.1.3 On completion of high temperature operating test and while still at the operating temperature, the insulation resistance shall be 20 megohms minimum.

3.3.1.6.1.1.4 On completion of high temperature operating test and within 1 hour of returning to ambient temperature, the insulation resistance shall be 100 megohms minimum.

3.3.1.6.2 AC circuits. All ac circuits of the inverter must be capable of withstanding 1250 volts rms, 60 Hz to the frame, with the dc circuit grounded to the frame, for a period of at least 1 minute without any arcing or breakdown of insulation.

3.3.1.6.2.1 The insulation resistance at 500 volts dc shall be 100 megohms minimum between all inverter ac circuits and the frame, with all dc circuits grounded to the frame.

3.3.1.6.2.1.1 Within 3 hours of completion of the temperature and humidity nonoperating test, the insulation resistance shall be 1 megohm minimum.

3.3.1.6.2.1.2 Within 48 hours of completion of the temperature and humidity nonoperating test, the insulation resistance shall be 100 megohms minimum.

3.3.1.6.2.1.3 On completion of high temperature operating test and while still at the operating temperature, the insulation resistance shall be 20 megohms minimum.

3.3.1.6.2.1.4 On completion of high temperature operating test and within 1 hour of returning to ambient temperature, the insulation resistance shall be 100 megohms minimum.

3.3.1.7 Running time. The maximum continuous running time of the inverter shall be 90 minutes, followed by sufficient time for the unit to cool to ambient temperature.

3.3.1.8 Start-stop life. The inverter shall be capable of withstanding a minimum of 1000 start-stop operations during its life in addition to the start-stop operations encountered during testing to this specification.

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3.3.1.9 Operating life. The inverter shall have a useful operating life of at least 200 hours without maintenance, except as specified in 3.3.1.3.2, in addition to the operating hours encountered during testing to this specification.

3.3.1.10 Over-voltage. The inverter shall be capable of withstanding an input over-voltage stress of 35 volts dc for 30 plus or minus 5 seconds.

3.3.1.11 Transient voltage. At no time when the inverter is operated in accordance with this specification shall the inverter generate a transient voltage, measured as near the inverter terminals as possible, that exceeds 35 volts peak.

3.3.1.12 Vibration. The amplitude of vibration, when measured approximately at the area of the 2-7/8 maximum center line and the 12.000 basic dimension locating the mounting holes on the mounting plate, as shown on Dwg 1263725, shall not exceed 0.0005 inch peak-to-peak when the inverter is operating in accordance with 3.3.1.2.2.3.

3.3.2 Product characteristics

3.3.2.1 Voltage setting. The voltage setting shall be capable of adjusting the output voltage plus or minus 5 percent or more, in increments no greater than 1 percent of the nominal value.

3.3.2.1.1 Method. The voltage setting shall be externally accessible from the plug end of the inverter control box, and shall be operable either manually or by means of a straight screwdriver. The setting shall be so designed that it will not be displaced by vibration or shock under the conditions specified in 3.4.

3.3.2.2 Frequency setting. The frequency setting shall be capable of adjusting the output frequency plus or minus 2.5 percent or more, in increments no greater than 1 percent of the nominal value. Adjustment will be required if the frequency drifts more than 2 percent during 200 hours of useful inverter life.

3.3.2.2.1 Method. The frequency setting shall be externally accessible and shall be operable either manually or by means of a straight screwdriver. The frequency setting shall be designed so as not to be displaced by vibration or shock under conditions specified in 3.4.

3.3.2.3 Voltage sensing. The voltage regulator shall sense at-least the AB phase of the inverter output as defined in 3.3.2.4.3.

3.3.2.4 Internal connections

3.3.2.4.1 Commons. There shall be no electrical connections made between the dc and ac circuits within the inverter.

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3.3.2.4.2 Grounds. No circuit within the inverter shall be connected to the inverter frame.

3.3.2.4.3 Receptacle. The inverter control box shall mount a receptacle, MS-3102-A-20-22P, as shown in Dwg 1263725. Receptacle pins shall be utilized as follows:

Pin A - plus 26 volts dc
Pin B - Line B, output voltage
Pin C - minus 26 volts dc, running power
Pin D - Line A, output voltage
Pin E - Spare
Pin F - Line C, output voltage

3.3.2.5 Brush replacement. Brushes shall be accessible for replacement when the brush opening covers are removed.

3.3.2.6 Schematic diagram. Attached to the inside of the control box shall be a complete circuit diagram describing all the electrical component values of the equipment. The diagram shall be legible, protected against humidity, and not easily removed.

3.3.2.7 Setting-means identification. The location of the output voltage and frequency setting device shall be identified on the inverter. Marking shall be legible, protected against humidity, and not easily removed.

3.3.2.8 The conducted and radiated interference when tested over the AN/ARM-6 Range (14 to 250 kHz) and as prescribed in 4.4.8 shall not exceed the limits given in Figures 1 and 2.

3.4 Environmental requirements. The inverter shall meet the other applicable requirements of this specification during or after the tests defined or specified in 4.5.

3.4.1 Vibration. The inverter shall meet the following vibration requirements:

3.4.1.1 Vibration, nonoperating. The inverter shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.1.2.

3.4.1.2 Vibration, operating. The inverter shall show no damage and shall meet all the performance requirements and product characteristics of this specification during and after the test specified in 4.5.1.3.

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3.4.2 Temperature and humidity

3.4.2.1 High temperature and humidity, nonoperating. The inverter shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.2.1.

3.4.2.2 High temperature, operating. The inverter shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.5.2.2.

3.4.2.3 Low temperature, nonoperating. The inverter shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.2.3.

3.4.2.4 Low temperature, operating. The inverter shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.5.2.4.

3.5 Cleanup. Prior to and after final assembly, all parts, components, and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges, as well as rosin flash, shall be removed.

3.6 Workmanship. The equipment, including all parts and accessories, shall be constructed and finished in a reamer to ensure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, riveting, machine screw assemblage, welding, brazing, and freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. Inspection of the inverter shall be classified as follows:

(a) Reproduction inspection (paragraph 4.5)

(b) Quality conformance inspection (paragraph 4.4)

(c) Periodic production inspection (paragraph 4.5)

4.3 Sampling

4.3.1 Preproduction. A preproduction sample of five inverters, manufactured in accordance with 3.2.3, shall be subjected to the preproduction tests detailed in 4.5 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the inverters by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Lot size. Unless otherwise specified in the contract or order, each shipment of units of product shall be a separate “inspection lot” and may differ from the quantity specified in the total order or contract. In any event, all units of a single "inspection lot” shall be made to the same design, with the same materials and manufacturing techniques, and shall be part of one continuous production run.

4.3.3 Periodic production samples. From each succeeding lot or grand lot of 300 or less that have passed the quality conformance inspection of 4.4, a random sample of five inverters will be selected for submission to an activity designated by the procuring activity to determine compliance of the samples with the requirements of the contract, specification, and drawings. The inverters shall be considered satisfactory if they meet all specifications and pertinent drawings requirements and withstand the tests specified in 4.4 and 4.5. Failure of the periodic production sample to comply with these requirements will result in the rejection of the lot, as determined by the procuring activity.

4.4 Quality conformance inspection. Each inverter delivered under contract shall be subjected to the following tests and examinations. Failure to meet any requirement specified herein shall be considered cause for rejection.

4.4.1 Test conditions. Unless otherwise specified, the inverter shall be subjected to quality conformance tests under the following conditions:

4.4.1.1 Temperature: Room ambient 65 degrees Fahrenheit degrees F (18.33 degrees centigrade (C) to 35 degrees C).

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4.4.1.2 Altitude: Normal ground.

4.4.1.3 Vibration: None.

4.4.1.4 Humidity: Room ambient to 95 percent relative maximum.

4.4.1.5 Ambient interference level. It is desirable that the ambient interference level during testing be at least 6 db below the allowable specified interference limit. However, if the levels of ambient interference plus test item interference are not above the specified limit, such tested item shall be considered in accordance with the requirements of this specification.

4.4.1.5.1 Frequency measurements. Measurements shall be made at not less than 3 frequencies per octave. The frequencies selected shall be chosen to avoid crowding of readings at one end of the band and at calibrated points. All other frequencies shall be scanned while monitoring with headphones or a loudspeaker. If interference peaks appear while scanning, measurements shall be made at each frequency where such peaks occur.

4.4.1.5.2 Interference instrument calibration. The standard calibration signal shall be the rms voltage of an unmodulated sine wave. The field intensity measuring instrument shall be calibrated using a standard signal generator and in accordance with the procedures given in the instruction manual of the interference meter. The time interval between calibration checks shall never exceed 6 months. Recalibration of the field intensity meter and the standard signal generator shall be made whenever checks indicate a deviation greater than 10 percent from the normal condition.

4.4.2 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel for perforating all quality conformance tests. The test equipment shall be adequate in quantity and, when definite requirements are not specified, shall be of sufficient accuracy and quality to permit performance of the required quality conformance tests.

4.4.3 Test equipment. The following items of test equipment are required to perform the quality conformance tests set forth in this specification:

4.4.3.1 DC power supply

4.4.3.1.1 Voltage: DC voltage adjustable from 10 to 35 volts under load.

4.4.3.1.2 Capacity: 40 amperes steady state minimum and 300 amperes surge minimum.

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4.4.3.1.3 Minimum voltage during surge: 22 volts dc at 300 amperes peak at 28 volts dc input.

4.4.3.1.4 Impedance: 0.02 ohm maximum

4.4.3.2 Voltmeters

4.403.2.1 DC: 0-50 volts dc, minimum resistance 100 ohms with an accuracy of 1 percent.

4.4.3.2.2 AC: 0-150 volts rms at 400 Hz, minimum impedance 3000 ohms, with an accuracy of ¼ percent.

4.4.3.3 Ammeter: 0-50 amperes dc, with an accuracy of 1 percent.

4.4.3.4 Wattmeters: Two single phase wattmeters are required with an accuracy of 1 percent.

4.4.3.4.1 Voltage circuit: 0-100 volts rms at 400 Hz, maximum burden 4 volts amperes full scale.

4.4.3.4.2 Current circuit: 0-5 amperes at 400 Hz, maximum burden 1 volt ampere full scale.

4.4.3.5 Electronic frequency counter: capable of measuring the operating frequency within 0.1 Hz.

4.4.3.6 Phase sequence indicator: The phase sequence indicator shall be made up of two 3 watt, 115 volt incandescent lamps, and a 0.06 microfarad, 200 volt dc capacitor, connected in wye.

4.4.3.6.1 Operation. The capacitor terminal shall be connected to line B; the ABC sequence indicating lamp shall be connected to line A; and the ACB indicating lamp shall be connected to line C, as designated in 3.3.2.4.3. See Figure 3. The phase sequence indicator circuit shall not be left in the circuit after phase sequence is determined or while other measurements are being performed.

4.4.3.7 Calibrated oscilloscope

4.4.3.7.1 Range of measurement: 1, 10, 100 volts full scale, with a multiplier of from 1 to 10 for each scale.

4.4.3.7.2 Frequency response: Flat to 500 Hz minimum.

4.4.3.7.3 Sweep: Recurrent, 5 to 500 Hz minimum

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4.4.3.7.4 Imput impedance: 1 megohm shunted by 50 micromicrofarads minimum.

4.4.3.8 High potential test set: The high potential test set shall be capable of furnishing 500 volts rms and 1250 volts rms at 60 Hz. The test set will be used for insulation testing and shall include an alarm bell or light, or a combination of both, to indicate when any arcing or breakdown of insulation occurs.

4.4.3.9 Dummy load: Three each of the loads described are required to make up the three phase delta inverter load. One resistive leg and one reactive-resistive leg are to be connected in parallel for each phase load. See Figure 4.

4.4.3.9.1 Resistive leg: A resistor or combination of resistors sufficient to provide 298 plus or minus 5 ohms total, with a tap point at 188 plus or minus 2.5 ohms. Maximum leg current: 0.65 ampere rms.

4.4.3.9.2 Reactive-resistive leg: The reactive-resistive leg shall consist of inductance and resistance in series. The resistive element may or may not be entirely made up of the inherent resistance of the inductor coil. Total inductance of 54.5 plus or minus 1 millihenries, with a tap point at 34.5 plus or minus 0.5 millihenries. Total resistance 45 plus or minus 1 ohms, with a tap point at 28.5 plus or minus 0.5 ohms. The reactive-resistive leg shall be constructed so that the two tap points may be used simultaneously. Maximum leg current: 1.32 amperes rms at 400 Hz.

4.4.3.10 Current shunt: A four terminal resistor (two current terminals and two voltage terminals) of 0.003 to 0,01 ohm having an accuracy of 1 percent and capable of carrying 50 amps continuous.

4.4.3.11 Interference measuring device: Stoddard, Model N17-10A or equivalent.

4.4.3.12 Amplitude of vibration measuring equipment: capable of measuring 0.0005 inch peak-to-peak or less.

4.4.3.13 Resilient ribbed neoprene pad: 5/16 inch thick, Isomode Pad, manufactured by MB Electronics, or equal.

4.4.4 Adjustment. Initial adjustment of output voltage and frequency shall be made as specified below. No additional adjustments of output voltage and frequency shall be made unless specifically permitted by the applicable test paragraphs (See 3.3.1.3.2 and 3.3.1.3.3).

4.4.4.1 Connect the tapped parts of the 3-phase delta dummy load of 4.4.3.9 to the inverter output terminals.

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4.4.4.2 Apply a source voltage of 26 plus or minus 2 volts dc to the inverter. After it is operating in the steady state condition, adjust the source voltage to obtain a voltage of 24 plus or minus 0.2 volts dc measured as near the inverter terminals as possible.

4.4.4.3 Adjust the inverter phase AB output voltage to 115 plus or minus 0.5 volts rms in accordance with 3.3.2.1.1.

4.4.4.4 Adjust the inverter output frequency to 400 plus or minus 4 Hz in accordance with 3.3.2.2.1.

4.4.4.5 If necessary, readjust the voltage at the inverter terminals to 24 volts dc, and recheck the output voltage and frequency settings to see that they conform to 4.4.4.3 and 4.4.4.4. If positive locks are provided on the voltage and frequency adjustment, tighten them securely.

4.4.5 Test procedure. See 3.3.2.4.3 for inverter terminal designations. The 98 plus or minus 2 percent rated load is obtained by using the tapped parts of the 4.4.3.9 dummy load; the 62 plus or minus 2 percent of rated load is obtained by using the entire 4.4.3.9 dummy load. The circuit impedance shall be as described in 3.3.1.1.2.

4.4.5.1 Starting conditions

4.4.5.1.1 Increasing voltage. Start the inverter as specified in 3.3.1.2.1. Perform and record the following measurements, using equipment as specified in 4.4.3.

4.4.5.1.1.1 Regulation. When a steady state voltage of 26 plus or minus 2 volts dc is attained, the output voltage measured at the terminals shall be within the limits of 3.3.1.3.2 and the output frequency shall be within the limits of 3.3.1.3.3.

4.4.5.1.1.2 Peak voltage. At no time during and after the starting conditions of 4.4.5.1.1 shall the peak phase AB voltage, as measured at the inverter output terminals with oscilloscope of 4.4.3.7, exceed plus or minus 230 volts.

4.4.5.1.1.3 Transient voltage. At no time during and after the starting conditions of 4.4.5.1.1 shall the dc input voltage, measured as near the inverter input terminals as possible, exceed 35 volts peak.

4.4.5.1.2 Step voltage. For the starting conditions of 3.3.1.2.2, measure and record the followimg:

4.4.5.1.2.1 DC input voltage. Measure and record the input voltage as near the inverter terminals as possible, using a 0-50 volt dc voltmeter (4.4.3.2.1).

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4.4.5 .1.2.2 DC input current. Measure and record the input current, using a 0-50 ampere dc ammeter (4.4.3.3).

4.4.5.1.2.3 AC output voltage. Measure and record the three ac output line-to-line voltages as near the inverter terminals as possible, using a 0-150 volt rms voltmeter (4.4.3.2.2) calibrated at 400 Hz. These voltages shall meet the requirements of 3.3.1.3.2.

4.4.5.1.2.4 Output frequency. Measure and record the phase AB frequency, using the electronic frequency counter (4.4. 3.5). This frequency shall be within the limits of 3.3.1.3.3.

4.4.5.1.2.5 Total ac output watts. Use two wattmeters (4.4.3.4) with 0-100 volts rms, 400 Hz voltage circuits and 0-5 ampere, 400 Hz current circuits. Connect plus or minus voltage and current terminals together on each wattmeter. Place the current circuit of one wattmeter in output line A, and place the current circuit of the other wattmeter in output line B, with the plus or minus markings on the inverter side. Join the remaining voltage circuit terminals together and connect them to output line C. Add the wattmeter readings for total watts output. See Figure 5.

4.4.5.1.2.6 Efficiency. Efficiency shall be not less than the limits specified in 3.3.1.4 when calculated according to the following formula, using the data taken in 4.4.5.1.2.1, 4.4.5.1.2.2, and 4.4.5.1.2.5:

$$\text{Efficiency} = \frac{\text{Total ac output watts (100)}}{\text{(dc input voltage) (dc input current)}}$$

4.4.5.1.2.7 Modulation. Measure across the AB phase, using a calibrated oscilloscope (4.4.3.7) as follows:

4.4.5.1.2.7.1 Set the oscilloscope so that at least 25 plus or minus 5 Hz peak-to-peak of the 400 Hz output voltage can be seen. Use sufficient vertical gain so that the trace very nearly fills the screen. Using the oscilloscope calibrator, measure and record the maximum peak-to-peak voltage. This reading shall be called E max.

4.4.5.1.2.7.2 Adjust the trace vertical position control and the vertical gain control on the oscilloscope so that only about one quarter of the positive half of the waves can be seen on the screen. Using the oscilloscope calibrator, measure and record the maximum difference noted between the voltage peaks on the screen. This reading shall be called E. See Figure 6.

4.4.5.1.2.7.3 Modulation limit. Modulation shall not exceed the limit specified in 3.3.1.3.7 when calculated according to the following formula and using the data of 4.4.5.1.2.7.1 and 4.4.5.1.2.7.2.

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$$\text{Percent modulation} = \frac{\Delta E\ (100)}{E\ max - \Delta E}$$

4.4.5.1.2.8 Commutation. Observe the sparking which occurs between the motor commutator and the brushes. If only fine pinpoint sparking is noted, record the commutation as "satisfactory". If more than fine sparks are noted, record the commutation as unsatisfactory. For this test, remove the end bell so that the brushes can be observed.

4.4.5.1.3 Peak starting current and decay time. Connect the current shunt (4.4.3.10) in the input circuit and connect the oscilloscope (4.4.3.7) to the voltage terminals of the shunt. Start the inverter as specified in 3.3.1.2.2.3. The conditions of 3.3.1.1.3 shall be met, as determined by the oscilloscope reading.

4.4.5.2 Phase sequence. With the inverter operating under the load conditions specified in 3.3.1.2.2.2, connect the phase sequence indicator as specified in 4.4.3.6.1. Record the phase sequence as indicated by the lamp which burns brightest. The phase sequence indicator shall not be left in the circuit after phase sequence is determined or while performing other measurements. The removal of the phase sequence circuit will prevent interference with the rated load to the inverter. The indicated phase sequence shall be ABC.

4.4.5.3 Stop the inverter by shutting off its input power supply. Replace any end bells or covers previously removed and remove the load specified in 4.4.5.2.

4.4.5.4 Dielectric. Apply test voltages to the inverter terminals, as specified in 3.3.1.6. Record any arcing or insulation breakdown that occurs.

4.4.5.5 Vibration. The inverter shall be placed on a pad (4.4.3.13) that is mounted on a hard flat surface. The inverter shall meet the requirement of 3.3.1.1.1.

4.4.6 Certification. The manufacturer will certify that the requirements of 3.2.4.8 and 3.2.4.9 have been satisfied.

4.4.7 Visual inspection

4.4.7.1 The inverter shall conform to the requirements of 3.3.2.4, 3.3.2.5, and 3.3.2.7.

4.4.7.2 The inverter shall conform to the requirement of 3.3.2.6.

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4.4.8 Conducted interference

4.4.8.1 The inverter shall be tested for sine wave and random noise interference.

4.4.8.2 The frequency shall be scanned from 14 to 250 kHz, and measurements shall be made at each frequency at which peaks occur.

4.4.8.3 Inverter hookup for conducted interference test (Figure 7)

4.4.8.3.1 The plus side of the dc supply shall be connected to the ground terminal of the stabilizer.

4.4.8.3.2 The minus side of the dc supply shall be connected to the shielded power source connecter of the stabilizer.

4.4.8.3.3 The shielded sample connector of the stabilizer shall be connected to terminal C of the inverter connector.

4.4.8.3.4 Terminal A of the inverter connector shall be connected to the ground terminal of the stabilizer.

4.4.8.3.5 The mounting base of the inverter shall be electrically connected to the ground terminal of the stabilizer and the stabilizer connected to earth ground.

4.4.8.3.6 The impedance stabilization network shown on Figure 8 shall be used. This device shall be connected between the load and the inverter and these components shall all be fitted with a series N, 50 ohm connects, to which the interference measuring instrument cable is attached for conducted interference measurements. This network shall have impedance versus frequency characteristics within 20 percent of the values shown on Figure 9.

4.4.8.3.7 The inverter shall be tested when in the condition described in 4.4.4.1, 4.4.4.2, 4.4.4.3, and 4.4.4.4.

4.4.8.4 Determination of interference type. In general, electromagnetic interference will be considered to be CW when it is sharply tunable on the field intensity meter, nontunable interference being considered to be broad band. For the purpose of this specification, CW signals will be defined as those signals for which the ratio of the peak meter indication to the average or field strength meter indication gives a value of 2.8 or less. Signals for which this ratio is greater than 2.8 are broadband.

4.4.8.4.1 Interference type and measurement method. For the measurement of broadband electromagnetic interference, the impulse generator substitution method with either a slideback peak detector or a direct reading peak-detector shall be used.

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4.4.8.4.1.1 CW electromagnetic interference shall be measured using an average detector. The method employed shall be either the substitution of an unmodulated sine wave or by precalibration of the field intensity meter using a sine wave generator.

4.4.8.4.1.2 Impulsive electromagnetic interference shall be measured using a slideback peak detector up to 30 mHz for repetition rates less than 10 pulses per second (pps) and above 30 mHz for repetition rates less than 100 pps.

4.4.9 Visual and dimensional examinations. Each inverter shall be examined to determine conformance with the requirements for workmanship and dimensions.

4.4.10 Packaging, packing, and marking. Packaging, packing, and marking shall comply with the requirements of Section 5.

4.5 Reproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed in 4.4, shall be subjected to the following tests and examinations. Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the procuring activity.

4.5.1 Vibration

4.5.1.1 Mounting. The inverter, less pack or packaging, shall be mounted rigidly to a vibration device by means of the mounting holes shown in Dwg 1263725 and tested as specified in 4.5.1.2 and 4.5.1.3.

4.5.1.2 Vibration, nonoperating. The inverter shall be subjected to the "Vibration, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.4.1.1.

4.5.1.3 Vibration, operating. The inverter shall be subjected to the "Vibration, operating" test of MIL-E-82590 and during and thereafter shall meet the requirements of 3.4.1.2.

4.5.2 Temperature and humidity

4.5.2.1 High temperature and humidity, nonoperating. The inverter shall be subjected to the "High temperature and humidity, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.4.2.1.

4.5.2.2 High temperature, operating. The inverter shall be subjected

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to the "High temperature, operating" test of MIL-E-82590 and, while at the test temperature, shall meet the requirements of 3.4.2.2.

4.5.2.3 Low-temperature, nonoperating. The inverter shall be subjected to the "Low temperature, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.4.2.3.

4.5.2.4 Low temperature, operating. The inverter shall be subjected to the "Low temperature, operating" test of MIL-E-82590 and, while at the test temperature, shall meet the requirements of 3.4.2.4.

4.5.3 Operating life. Start and adjust the inverter as specified in 4.4.4. Operate the inverter for a total of 199 hours plus or minus 1 hour, maintaining the stead state input voltage (measured as near the inverter terminals as possible) at 24 plus or minus 2 volts dc. At the end of each 85 plus or minus 5 minutes of operation, stop the inverter. The inverter may be allowed to cool to ambient temperature before it is again started and operated as described above. Once during each 10 hours of operation the voltage and frequency setting may be changed to ensure proper operation during the next 10 hour operating interval. During this test, no maintenance other than that specified above shall be required. While the inverter is operating in a steady state condition, performance shall be within the limits specified in 3.3.l.3.2 and 3.3.1.3,3. At the conclusion of this test, the inverter shall pass all the quality conformance tests of 4.4.

4.5.4 Response time

4.5.4.1 Increasing voltage. Start the inverter as specified in 3.3.1.2.1. Response time shall be measured using suitable calibrated equipment (such as an oscillograph in conjunction with a frequency-to-voltage converter) and shall meet the requirements of 3.3.1.3.8.2.

4.5.4.2 Step-voltage. Start the inverter as specified under the conditions of 3.3.1.2.2. Response time shall be measured with suitable calibrated equipment (such as an oscillograph in conjunction with a frequency-to-voltage converter) and shall meet the requirements of 3.3.1.3.8.1.

4.5.5 Over voltage. Apply an open circuit source voltage of 35 plus 2 minus 0 volts dc to the inverter with 62 plus or minus 2 percent of rated load connected. Within 30 plus or minus 5 seconds after the inverter is started, reduce the source voltage to 26 plus or minus 2 volts dc. The inverter shall then meet the requirements of 3.3.1.3.2 and 3.3.1.3.3.

4.5.6 Load change recovery time. Start the inverter as specified in 3.3.1.2.2.2 and 3.3.1.2.2.4. For each of these starting conditions, switch the inverter output to a load of 98 plus or minus 2 percent of rated load.

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Recovery time shall be measured with suitable calibrated equipment (such as an oscillograph in conjunction with a frequency-to-voltage converter) and shall meet the requirements of 3.3.1.3.9.

4.5.7 Harmonic voltages. Start and operate the inverter as specified in 4.4.4.1, 4.4.4.2, 4.4.4.3, and 4.4.4.4. Using a wave analyzer connected across the AB phase, measure the individual odd harmonics up to and including the twenty-fifth. The harmonic voltages shall not exceed the limits specified in 3.3.1.3.6. Total harmonic content is defined as:

$$E_{total} = \sqrt{E_3^2 + E_5^2 + E_7^2 + \ldots\ldots\ldots + E_{25}^2}$$

where the subscript denotes the harmonic and all voltages are measured in the same units as the fundamental component.

4.5.8 Radiated interference

4.5.8.1 The inverter shall be tested for sine wave and random noise interference e.

4.5.8.2 The frequency shall be scanned from 14 to 250 kHz, and measurements shall be made at each frequency at which a peak occurs.

4.5.8.3 The inverter hookup and the determination of interference type shall be the same as given in 4.4.8.3 and 4.4.8.4 for the conducted interference test.

4.5.8.4 A 41-inch rod antenna shall be positioned vertically distance of 3 feet from the closest edge of the inverter. The inverter shall be placed on 3/4 inch plywood with a bond to a ground plane. Antennas shall be rotated around the inverter for maximum interference meter reading.

4.5.8.5 The inverter load during the test shall be the same as given in 4.4.8.3.7 for the conducted interference tests.

4.5.9 Start-stop life. Start the inverter as specified in 3.3.1.2.2.1. When the inverter has reached a steady state condition, as determined by the phase AB output voltage, shut off the inverter. When the inverter has reached a dead stop, repeat the foregoing steps. Repeat the above for 1000 Hz. The test maybe stopped and the inverter allowed to cool to ambient temperature at the end of each 85 plus or minus 5 minutes of cycling. Following the last cycle, the inverter shall meet the requirements of 3.3.3, 3.3.1.3.2, 3.3.1.3.2.1, and 3.3.1.3.3.

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5. PREPARATION FOR DELIVERY

5.1 Reservation and packaging. The levels of preservation and packaging shall be as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Unit packaging. The inverter shall be packaged in accordance with Method IID of MIL-P-116 and LD 495963.

5.1.2 Level B. Not applicable.

5.1.3 Level C

5.1.3.1 Unit packaging. The inverter shall be packaged in accordance with Method III of MIL-P-116.

5.2 Packing. Packing shall be Level A or Level C, as specified (see 6.2).

5.2.1 Number of packages per pack. The number of unit packages per pack, for Level C, shall be as specified in the contract or order.

5.2.2 Level A

5.2.2.1 Unit pack. The inverter shall be packed in accordance with LD 495963 and may be shipped individually in small lots.

5.2.2.2 Palletizing. When specified, Level A packs shall be palletized in accordance with MIL-STD-147, Pattern No. 90 (see 6.2).

5.2.3 Level B. Not applicable.

5.2.4 Level C. The inverter shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Marking

5.3.1 Normal markings, Level A. Unit packs shall be marked in accordance with LD 495963.

5.3.2 Normal markings, Level C. Unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

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5.3.3 Serial numbers. Unit packages and shipping containers shall be marked with the serial numbers of the units contained within.

6. NOTES

6.1 Intended use. The inverter is intended for use in Torpedo Mk 37 Mods 0, 2, and 3.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Any exceptions to this specification.

(c) Required level of preservation and packaging (see 5.1).

(d) Required level of packing, including palletizing (see 5.2).

(e) Assigned activity for preproduction and -periodic production inspections.

(f) Data required other than that specified in 3.1.1.

(g) That additional samples required because of failure of any unit(s) to pass the prescribed periodic production inspections shall be provided and inspected at the expense of the supplier.

Preparing Activity
Navy - OS
Project No. 1355-N253

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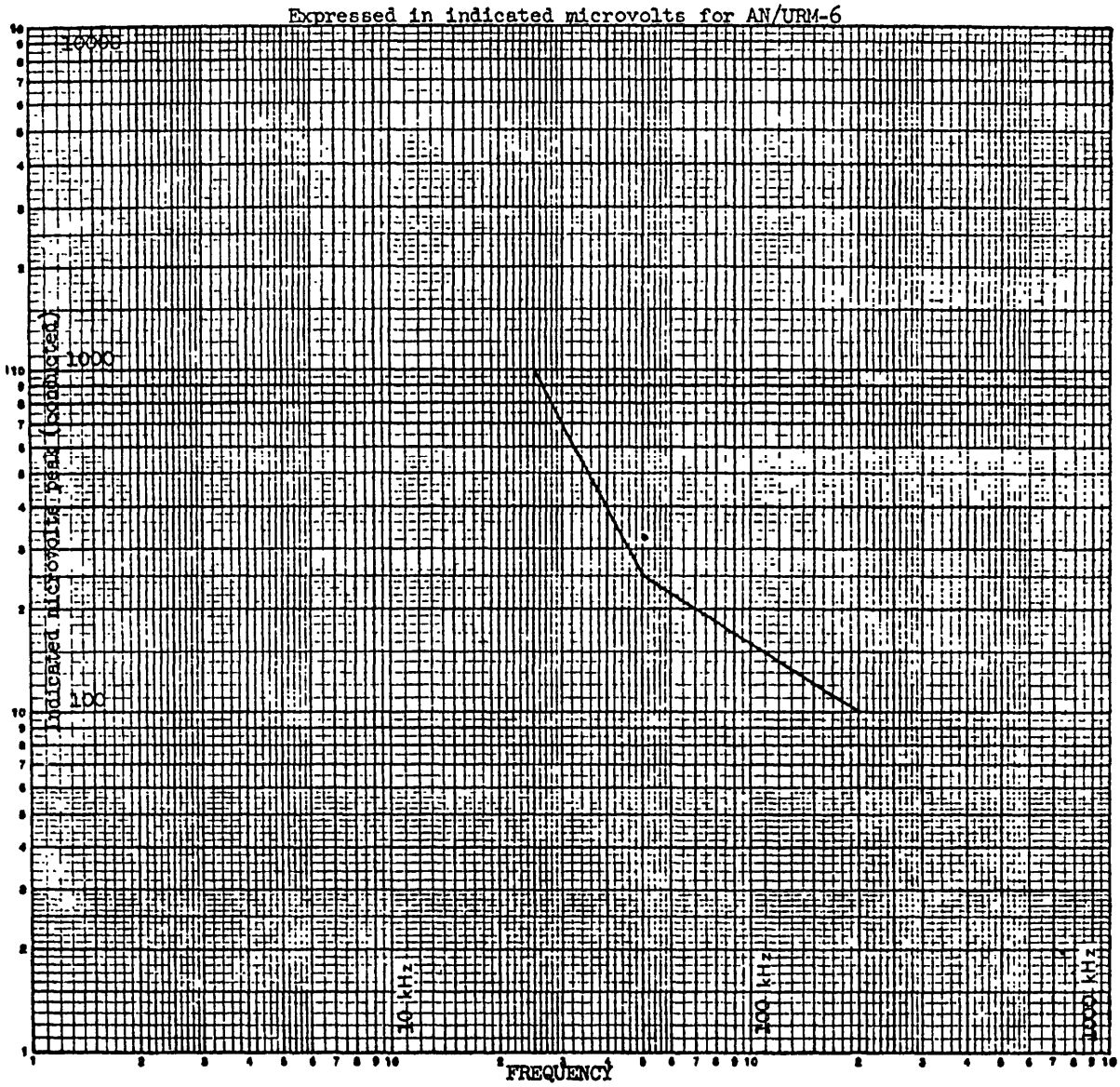


FIGURE 1. BROADBAND CONDUCTED INTERFERENCE LIMITS

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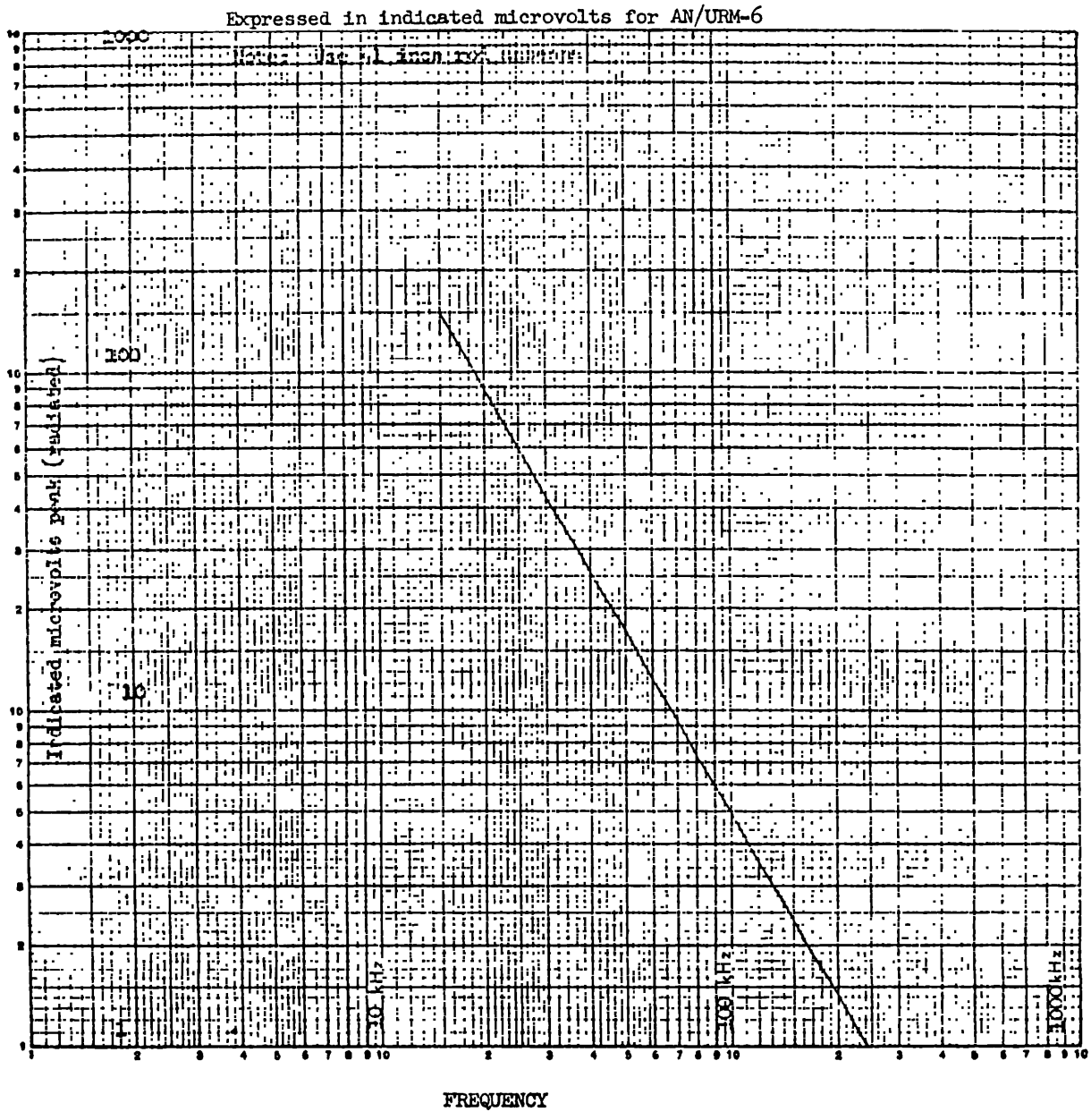


FIGURE 2. BROADBAND RADIATED INTERFERENCE LIMITS

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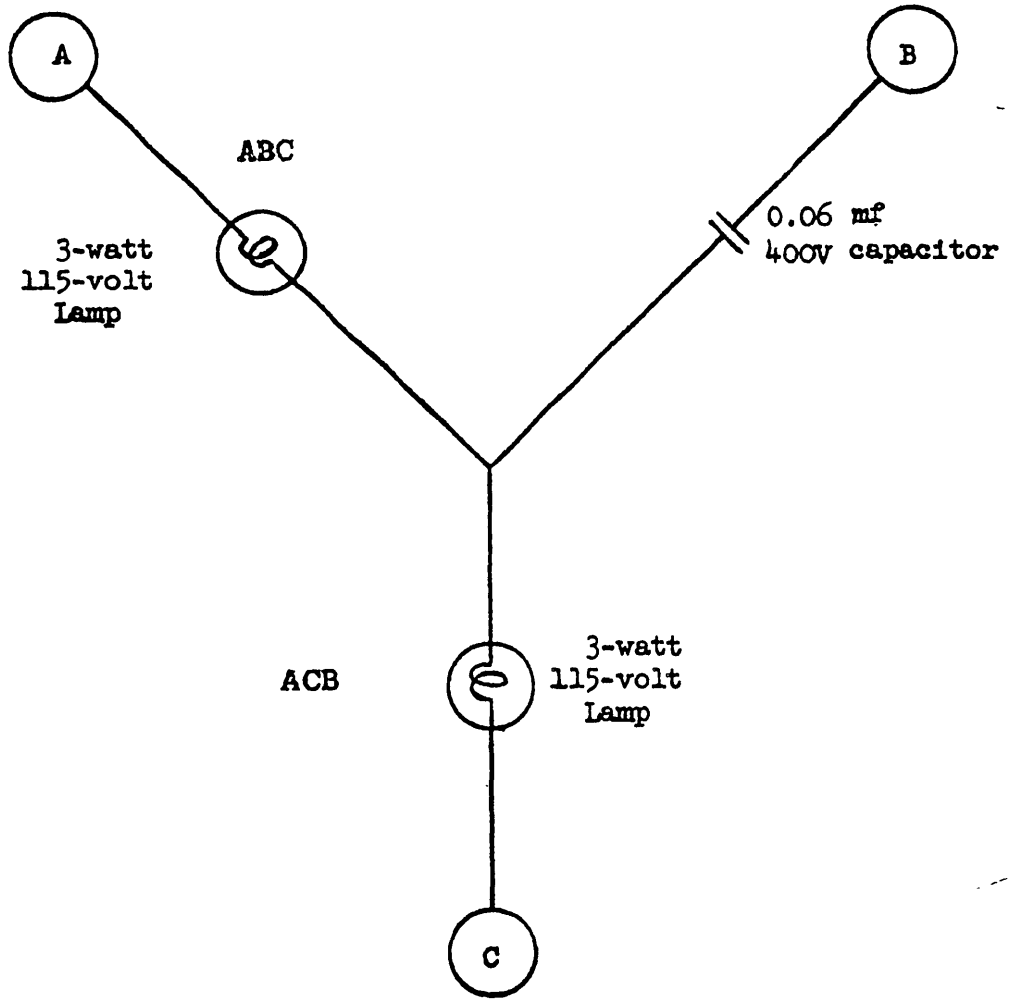
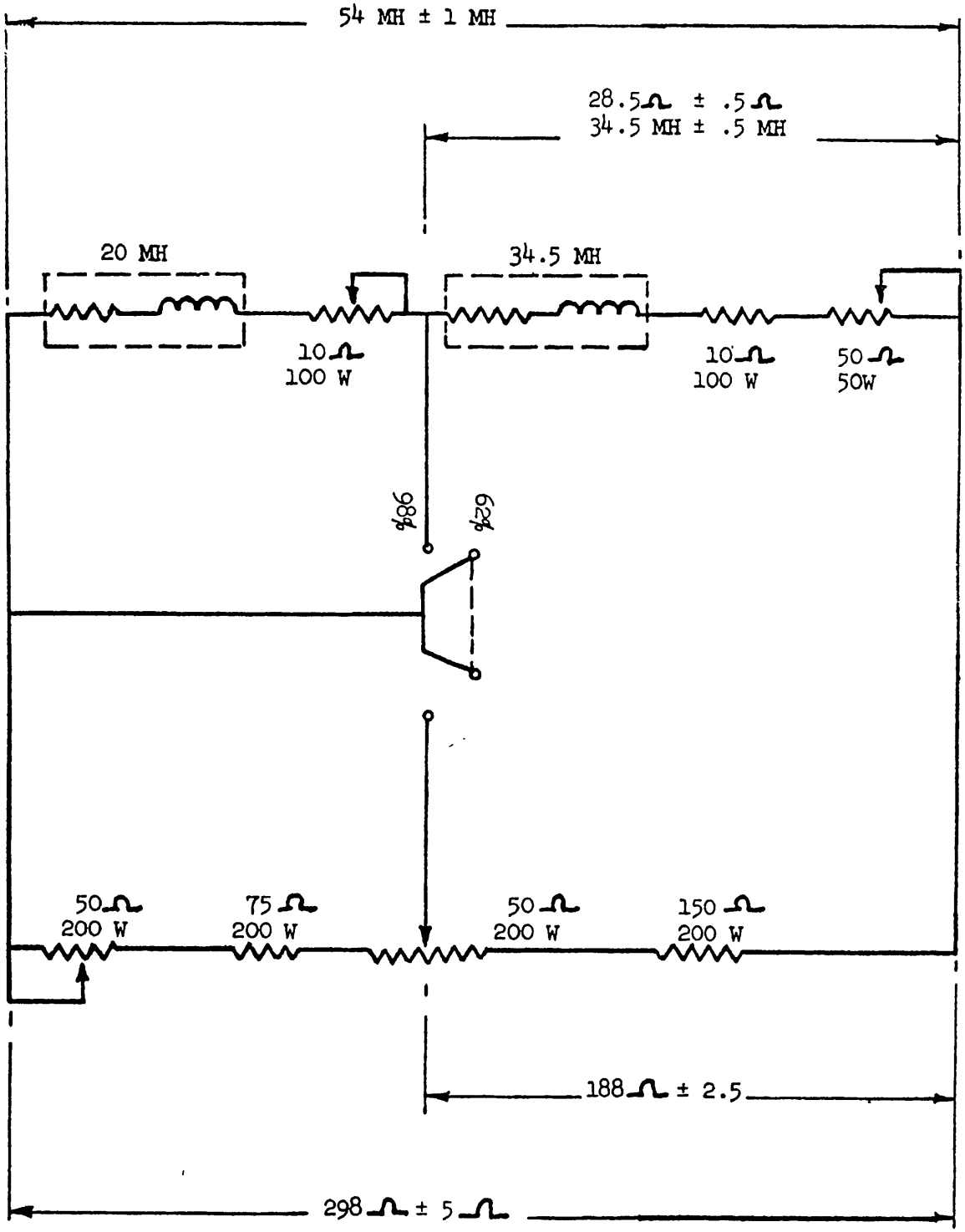


FIGURE 3. PHASE SEQUENCE INDICATOR

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RESISTIVE LEG

FIGURE 4. LOAD SCHEMATIC FOR EACH PHASE

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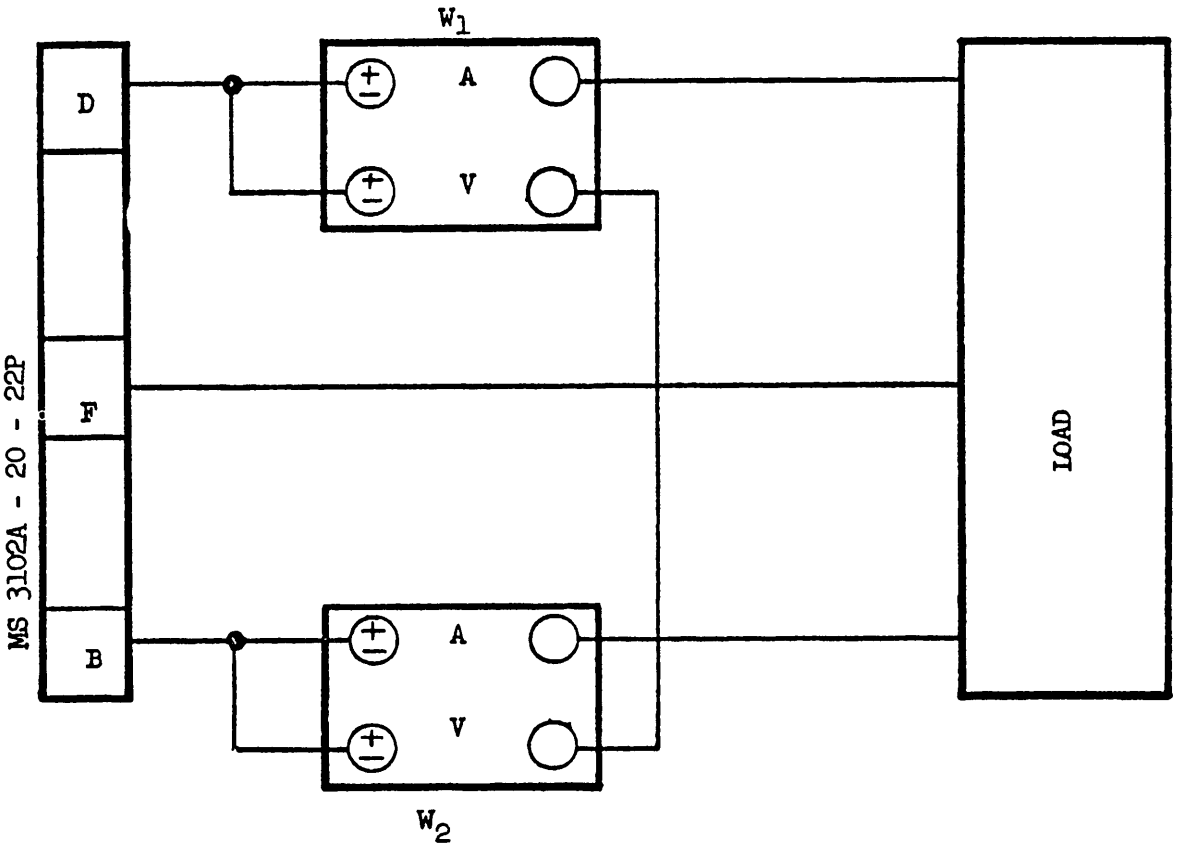


FIGURE 5. WATTMETER CONNECTIONS

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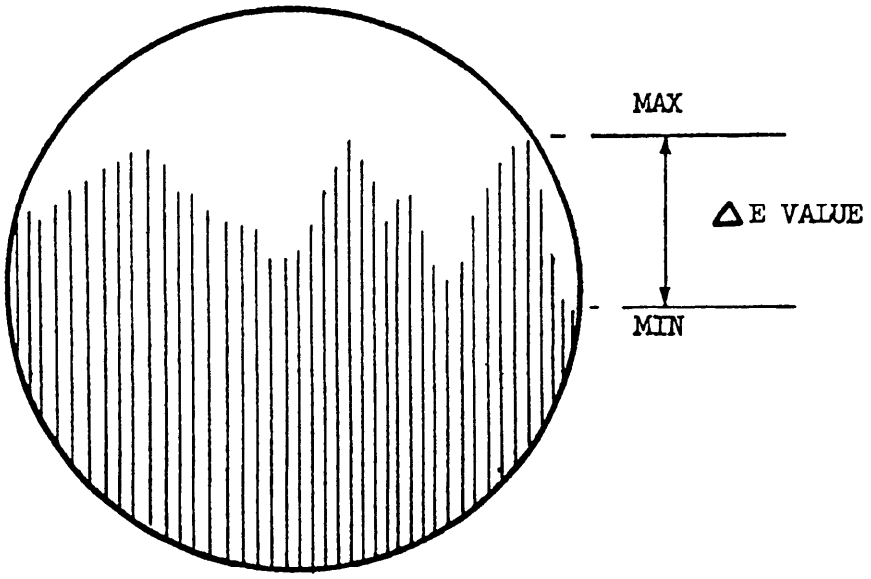


FIGURE 6. MODULATION PATTERN

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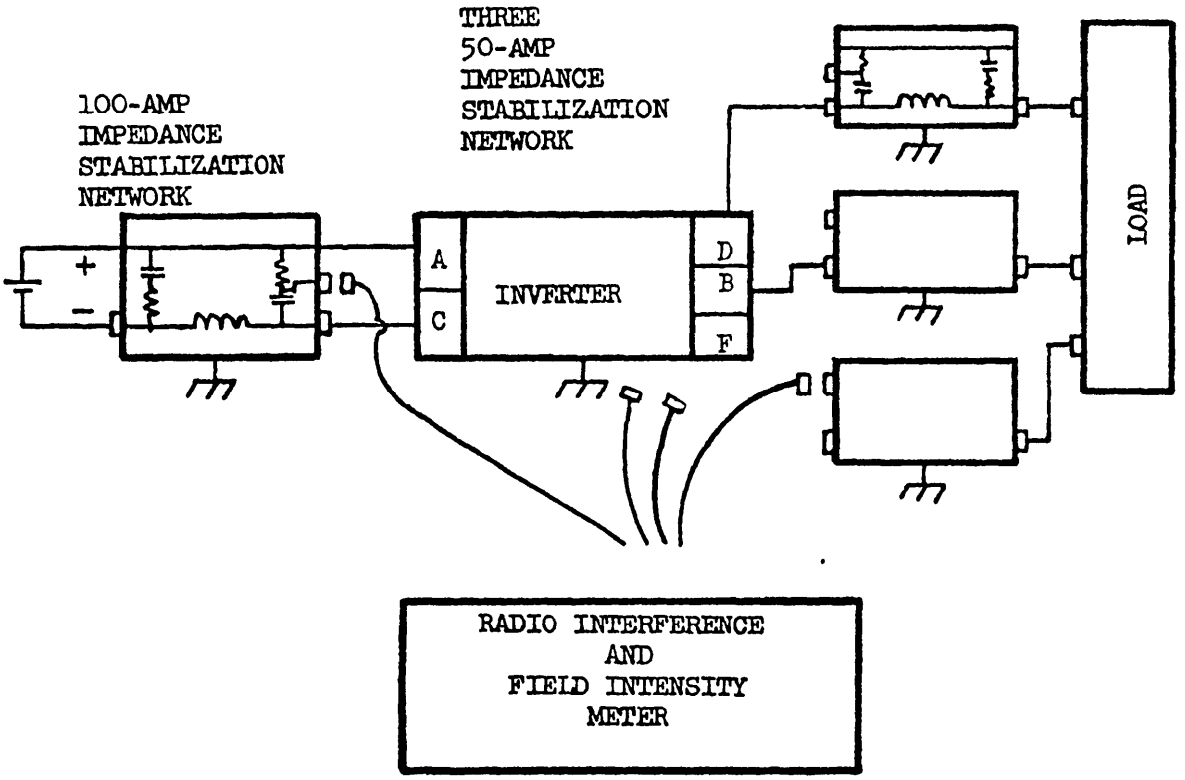
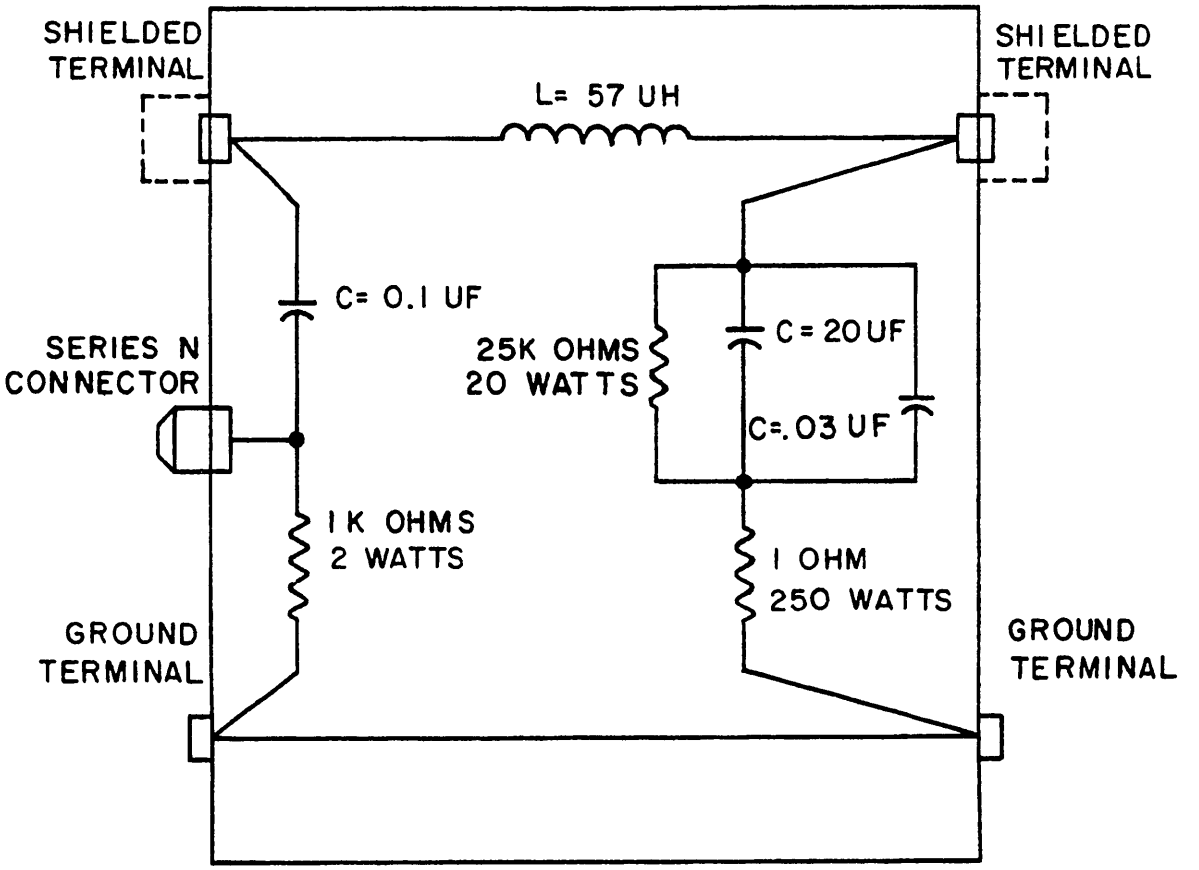


FIGURE 7

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COIL DATA:

1. L = 57 microhenrys at 15 kHz.
2. 26 turns No. 6 stranded wire, heavy insulated.
3. Plastic form 6 inches outer diameter.

METAL ENCLOSURE DIMENSIONS:

17½ inches x 9 inches x 9 inches.

FIGURE 8. LINE IMPEDANCE STABILIZATION NETWORK (14 kHz to 10 MHz)

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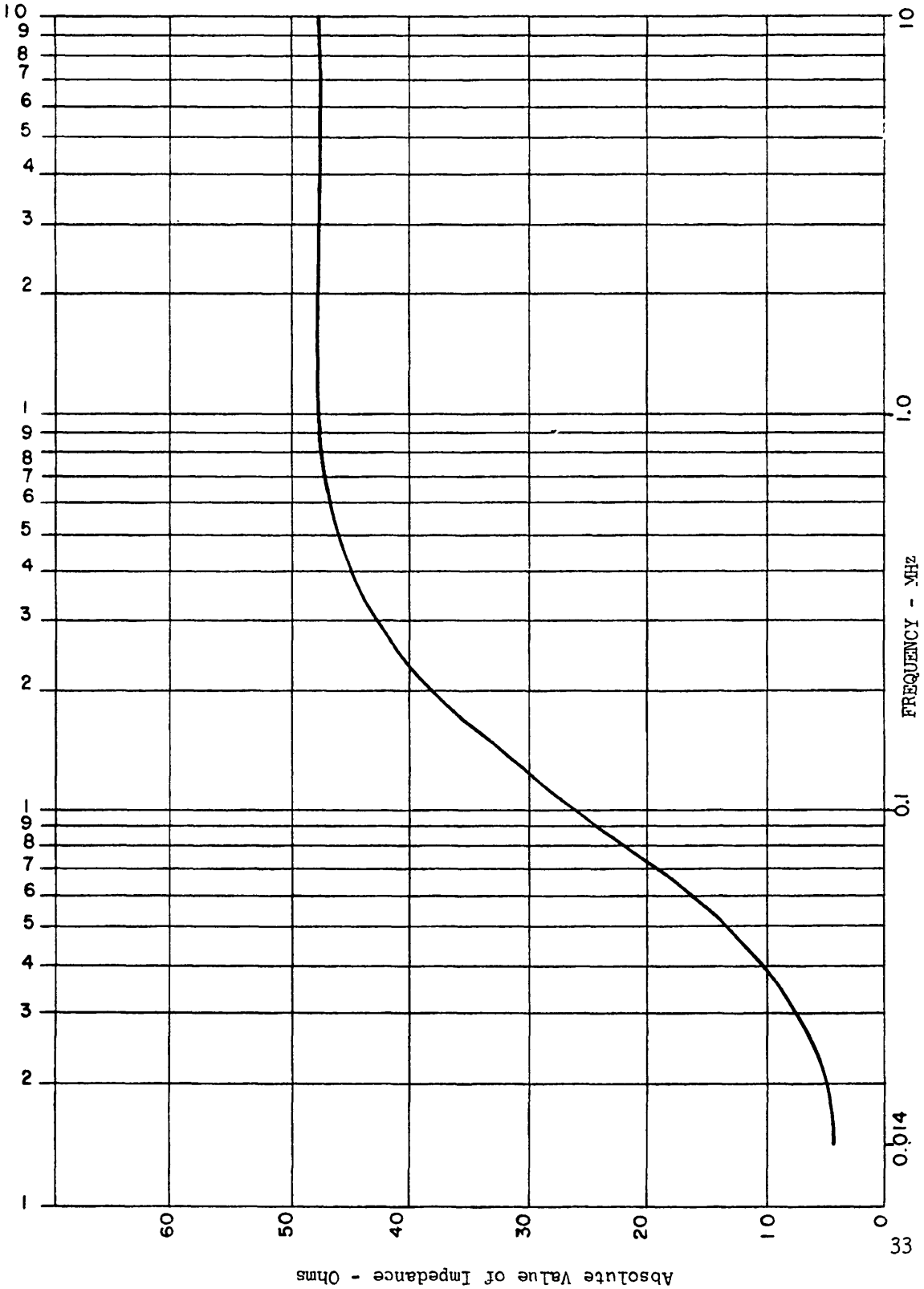


FIGURE 9. LOW FREQUENCY LINE IMPEDANCE STABILIZATION NETWORK

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